

Jayapriya P

bhuvaneshwari EEC (2)

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A Hierarchical Multi-Agent Deep Reinforcement Learning Framework for Privacy-Preserved and Low-Latency Cybersecurity in Cloud-Based Telemedicine Systems

Abstract—With the simultaneous explosion of connected medical devices, and increasing real-time patient monitoring capabilities, telemedicine infrastructures have witnessed a tremendous surge in demand for intelligent, adaptive, and privacy-preserving cybersecurity mechanisms. In fact, most of the currently available centralized Deep Reinforcement Learning (DRL) frameworks, that provides high detection accuracy, have high latency, scalability problem and less explainability. This paper proposes a new hybrid architecture HIER-MA-DRL (Hierarchical Multi-Agent Deep Reinforcement Learning) to overcome those limitations to facilitate distributed threat detection for edge/cloud environments. The proposed model maintains a balance between quick response for immediate anomalies and global situational observance by integrating lightweight edge agents but also using a centralized DRL coordinator for policy optimization over long-horizons. Moreover, embedded explainable policy distillation mechanisms to guarantee transparency and faithfulness in response to automation. We conduct extensive experimental evaluation using network intrusion benchmark datasets and real-time telemedicine traffic and demonstrate that HIER-MA-DRL achieves a detection accuracy of 99.03%, a response latency of 41.8% lower, and a false positive rate of 36.5% lower than that of the conventional centralized DRL frameworks. We ensure scalability with high data stream volumes due to our hierarchical coordination between agents and we protect patient data privacy using on-device anonymization. In summary, HIER-MA-DRL provides a strong, explainable, and low-latency cybersecurity framework that can be applied to next-generation smart healthcare and telemedicine environments.

Keywords—Hierarchical Multi-Agent DRL, Edge Computing, Cybersecurity, Telemedicine, Privacy Preservation, Explainable Reinforcement Learning, Low-Latency Detection, Cloud–Edge Architecture

I Introduction

Telemedicine and real-time patient monitoring have extremely changed the way healthcare is delivered, introducing a level of interconnectivity among medical devices, cloud and edge nodes on a level we have not seen before. On the other hand, the fast digitalization has enlarged the attack surface of health-care infrastructures and the need for intelligent cyber-security methods that can accommodate privacy concerns [1], [2]. Centralized security won't work because of latencies introduced processing high queued medical traffic, a common occurrence in tele-health applications [3][4], and doesn't also scale. While the combination of distributed learning paradigms and cloud–edge orchestration is an interesting direction to explore, upholding the desired level of speed in anomaly detection with compliance under confidentiality policies for highly sensitive patient environments are Current models may be robust and system-aware [5], but these do not consider hierarchical

mechanisms trading localized decision-making versus global optimisation. As a result, this creates an increasing demand for multi-layer, explainable, and compliant privacy-preserving learning systems able to meet the latency–security trade-off in cloud-based telemedicine environment [6]. Multi-agent reinforcement learning has also been applied as a disruptive adaptive cybersecurity and intelligent network control technology in the rapidly changing digital healthcare ecosystem [7]. Multi-agent systems enable distributed intelligence across heterogeneous healthcare networks by allowing agents to learn from changing contexts [8]. Nevertheless, traditional frameworks rely on centralized training, which introduces the risks of congestion and loss of exposures [9]. It is possible that the multi-agent learning can be embedded into a hierarchical control framework to enable localised anomaly suppression and global policy refinement or convergence. Such a concept is of paramount importance, particularly in latency-constrained applications as remote diagnostics and on-the-fly IoMT devices based on the continuous health monitoring of patients [10]. Furthermore, ensuring transparency and explainability of such learning processes has become a clinical demand and is an obligatory aspect in terms of medical data protection [11]. As such, a hierarchical learning ecosystem architecture including explainable deep reinforcement learning at both the edge and cloud layers can offer an additional antifragile cybersecurity defense in depth [12].

Recent works advocate hybrid cloud–edge deployment architecture in healthcare to reduce data flooding, increase data throughput and preserve privacy for distributed learning [13]. But despite these advances in technology, the telehealth systems are still facing challenges such as data sovereignty, reducing latency and flexible policy synchronizations between agents [14]. Privacy techniques activated at the edge (e.g., anonymization and differential learning) may bear positive potential of preserving patient data [15], however, these methods don't seem to be highly elastic in such a way when used over heterogeneous network infrastructure hindering the response time [16]. With the shift in complexity of healthcare workloads there is an unprecedented requirement to adopt automated analytics systems that continuously harness high performance computing to focus on prioritizing important security threats [17]. These discrepancies can be reconciled using a hierarchical reinforcement learning approach, while real time decision making is performed by the edge agents, which have been implemented to operate as cloud coordinators and are still able to optimize long term policies with experience sharing aggregation [18]. It considers this distributed yet coordinated setup to find the right balance between privacy protection, flexibility, and agility in telemedicine cybersecurity [19]. Cyber Security and telemedicine – Not just a technical problem, but an ethical duty for patient confidence, medical reliability and digital health evolution. Although classical DL-based ID models are often accurate, they are mainly black boxes that lack clinical interpretability and regulatory approval [20]. We integrate explainable reasoning and policy distillation components into our hierarchical multi-agent architecture, which as a result makes it possible for the system to exchange justifications of security decisions in order to obtain transparency in respect automated reactions to security threats. Furthermore, by using slim agents all over edge devices, latency-sensitive functionalities (e.g., anomaly filtering and intrusion flagging) can be performed in real time without sending all events to the cloud. On the other hand, a central coordinator can collect up policies and optimize decision rules to achieve better long term survival. Such a two-layer

orchestration can make telemedicine cybersecurity as a self-learning, privacy-preserving and latency sensitive ecosystem. To realize this, the present work develops a Hierarchical Multi-Agent Deep Reinforcement Learning (HIER-MA-DRL) model that enables cloud-edge intelligence and possesses explainability and privacy protection properties to enhance cybersecurity of next-generation telemedicine systems. The main contributions of this paper are:

- **H-MAC:** Hierarchical Multi-Agent Coordination Our hybrid DRL framework offloads the computational cost of optimal multi-agent coordination in hierarchical structure from edge agents to a central coordinator by taking advantage of local (short-term) threat mitigation responses and global (longterm), OPTimal security learning.
- **XPD Mechanism:** We use the explainable reinforcement learning to transform complicated policy updates into readable decision-rational in reasoning processes, making it more transparent and reliable of telemedicine cybersecurity system response.
- **Cloud-Edge Integration Combined with Privacy-Preservation & Latency-Optimisation:** Incorporates device-side anonymization and federated learning to trade-off between confidentiality of patient data in face of real-time intrusion response, versus scalability being maintained against a backdrop of large-scale medical data.

Organization of the paper: In Section II, we surveyed some related works focusing on privacy preservation, hierarchical intelligence, and DRL-based cybersecurity models in the telemedicine infrastructure. The proposed HIER-MA-DRL methodology is described in section III with its hierarchical agent architecture, policy, and explainable distillation. We detail the experimental design, dataset selection, performance metrics, and training parameters to validate detection accuracy and latency reduction in Section IV. The results and analysis, in Section V, compares the performance metrics such as detection rate, false positives, and computational latency of the proposed approach with centralized DRL frameworks. Section VI ends the paper with the main points and implications of the research and challenges for adaptive and privacy-aware cybersecurity in telemedicine.

II Related Works

Odeh et al. [1] proposed preserving the privacy of data in Industrial IoT infrastructures and proposed different methods such as anonymization, perturbation and differential privacy to secure data transmission in edge networks. They further demonstrated that, due to the vast dissimilarities in attack models in decidedly heterogeneous IoT settings, Dwork's differential privacy cannot at all be employed to twelve instances of industrial edge systems, and thus opened the avenue for hybrid privacy-preservation frameworks combining both cryptographic and anonymization based privacy-preservation measures towards security enhancements of industrial edge systems. Padmavathy (2021) [2] proposed a cloud-based telemedicine platform which improved scalability, data synchronization, and standard conformance with healthcare norms like HIPPA and GDPR. This system achieved enhanced interoperability with current electronic health records and a large reduction in latency

through real-time analytics, end-to-end encryption, and multi-factor authentication, thus strengthening the reliability of telemedicine communication channels. Kiran et al. (2025) emphasized cloud infrastructure scaling telehealth, giving examples from integrating I-IoT devices and federated cloud assets on lowering costs & increasing accessibility by tuning connectivity & enabling real-time healthcare delivery. The work also demonstrated the extent to which elastic cloud computing can converge patient, practitioner and healthcare system thanks to predictive analytics and common access to shared medical data repositories [3].

Hegde and Varughese et al.(2025) they introduced the Telecom-as-a-Service (TaaS) model to provide solutions for connectivity and security issues in hybrid cloud-IoT networks [4]. It incorporated an approach that provided managed security services, including real-time monitoring and endpoint protection, while benefiting from SDN-enabled multi-cloud environments to enhance network agility, resilience and compliance management across distributed healthcare systems. Moving forward, Thota (2024) [5] developed the blend of edge computing and artificial intelligence, in order to reduce the cloud latency for real-time workloads. We proposed a new way of deploying federated learning and lightweight AI models to the network edge for federated and efficient deployment to physical devices and described how the availability of even a few packet types can lead to much faster processing of traffic and a more reliable low-latency AI-driven workload manager enabling them to be more suited towards the inherently latency-sensitive features of a hyper-human healthcare environment. Jayanthiladevi et al. Dey et al. (2025) [6] proposed an AI-based cybersecurity framework specifically designed for 7G-enabled virtual therapy platforms, employing predictive analytics and adaptive threat intelligence to secure patient interaction. Their approach implemented a deep learning-based model that reduced the response time by 30% and the detection accuracy was 98.5%, which indicates that such a proactive AI-based defense would bring values to post-COVID-19 telehealth systems.

Cloud computing role on Industry5.0 goals developmental by holistic humanity structures' powered automated Intelligence [7] Patil (2025) Among the various cloud frameworks identified, those that shuttle microservice-based architectures that provision interoperability, ethical compliance and low-latency performance bear similarities in underlying principles to that of PV-published telemedicine infrastructures. IoT-Cloud integration enables real-time process optimization through predictive analytics and autocratic decision systems [8] as analysed by Ajayi (2015). This solution not only allows continuous monitoring of useful conditions but also minimizes operational latency and enhances dynamic resource allocation, which strengthens the advantages of efficiency and adaptability of data-intensive healthcare systems. Goriparthi(2024) [9] used reinforcement learning in IoT ecosystems by enabling devices to autonomously decide on actions to optimize energy consumption and respond to security challenges. Thus this approach through adaptive reinforcement learning based mechanism supported timely real-time detection of few anomalies, less dependency on centralized controllers thus supporting medical IoT deployments in latency-sensitive environments.

Naik et al. (2025) [10] proposed a hybrid deep learning approach with CNN, LSTM and VAE architectures under HIoT environments to improve the real-time anomaly detection and trust computation. Instead, a trust-aware controller by dynamically calculating reliability scores worked for their approach to maximize both access control and detection sensitivity with variable network conditions in medical IoT systems. Mulo et al. In 2025, [11] present a survey of deep learning applications in the Internet of Medical Things and introduce an IoMT framework that focuses on scalable and privacy-preserving IoMT models. The authors of the study suggested the application of transparent and explainable deep learning pipelines to guarantee strong as well as mobile medical data analysis in distributed health care networks. Ameen et al. (2024) A Novel Blockchain-based Cybersecurity Model For ECG Data in Internet of Medical Things Environments: Implementation and Application based on End to end Encrypted Communication and Feature Optimized SVM Classification [12]. The framework demonstrated a classification accuracy of 99.9%, indicating blockchain can ensure integrity and traceability in the process of biomedical signal processing. A quantum-enhanced digital twin Internet-of-Things framework that combines AI and quantum computing for healthcare task offloading was proposed by [13] Jameil and Al-Raweshidy (2025). The DTH-ATB-MAPPO and AQDT-IoT outperformed other methods in achieving 32% optimized offloading performance and lowered latency, paving the way towards quantum-enhanced, secure, and scalable telemedicine systems.

Moqurab et al. (2022) [14], a fog-based privacy-preserving healthcare framework is discussed with a focus on using the CNN-BiLSTM networks for the real-time entity extraction. In terms of high precision and recall (over 92 %) as well as low delay, this system demonstrates how effective fog computing is in achieving privacy-aware and responsive smart healthcare applications. Dhasarathan et al. Shahraki et al. (2024) [15] proposed a multi-agent privacy metrics framework based on an hybrid agent-based deep learning approach. This method provided dynamic privacy metrics—confidentiality, integrity, and availability—on the data access and at the same time, it distinguished the risk of unauthorized exposure of the data and increased the diagnostic accuracy. Saini et al.(2024) [16] developed a hybrid machine learning model that integrated k-anonymity, Firefly optimization, and time-aware allocation algorithms to strengthen trust and privacy in cloud environments. It maintains a mean with 93% accuracy and improves the overall performance of data resources through a guaranteed way of managing a secure balance between efficiency and confidentiality in distributed networks. Bose et al.(2025)[17] proposed a lightweight differential extreme gradient reinforcement neural network integrated with stochastic vector autoencoder based adversarial regression for anomaly detection in mobile edge computing systems. The detection accuracy of nearly 99% with 98% privacy protection, indicated the efficacy of this hybrid in enabling a scalable privacy-preserved approach to anomaly management in healthcare data streams.

2.1 Research Gap

While much progress has been made through heterogeneous frameworks and computing paradigms throughout IoT, cloud and edge-enabled healthcare systems [1]–[17], there still remain substantial gaps in the privacy-preserving, detection accuracy and response latency

triad. Existing solutions either tackle the problem in a centralized architecture that provides a high learning ability which comes with a high communication overhead and high latency in decisions [2], [3], or based on an edge lightweight framework that leads to a low detection accuracy due to light-weighting computations [5], [8]. While blockchain based [12] and multi-agent models [15] have provided significant improvements over existing systems in transparency and decentralization, at high volumes of telemedicine data streams they also introduce additional synchronization latency and scalability issues. Likewise, hybrid deep learning and reinforcement learning methods [9], [10], [17] demonstrate the potential for adaptive threat detection but also fail to incorporate hierarchical coordination mechanisms to ensure alignment of local edge solutions to global policies at the cloud-level. Additionally, in the context of DRL-driven cybersecurity explainability and interpretability have not been well studied, which not only hinders the trust of decision makers, but also regulatory conformance of medical environments [11]. One that combines edge-level responsive agent, cloud-level policy updating agent and privacy-preserving computing into a comprehensive hierarchical multi-agent DRL framework is thus required to deliver on-time, explainable and scalable cybersecurity for next-generation telemedicine infrastructures.

III Proposed Methodology

The HIER-MA-DRL approach in the proposed methodology exploits a hierarchical structure of three operational tiers—edge, and fog, and cloud—to realize adaptive privacy-preserved cyber-security in telemedicine environments [20]. At the edge tier, lightweight DRL (Deep Reinforcement Learning) agents conduct continuous device behaviour monitoring, also performing on-device anonymization, to safeguard patient data before it is transmitted. Fog layer serves as an interface to aggregate local threat intelligence and enable coordination between different agents in order to avoid alerts duplication and temporal consistency in responses [21]. At the same time, the cloud coordinator performs long-horizon policy optimization via deep Q-networks and actor-critic models, allowing for global knowledge and adjustment of local agent policies. To ensure that training can be distributed yet still efficient without compromising data privacy, we incorporate federate learning mechanisms into our method to allow training without revealing any raw data to make the model robust without sacrificing confidentiality. Privacy-preserving encryption, adaptive trust scoring, and edge offloading together enable a secure and efficient solution for telehealth environments [22]. Simulation environments mimic real telehealth traffic and agents are trained and evaluated on benchmark datasets for intrusion detection over multiple performance metrics such as accuracy, false positive rate, and response latency. Such an orchestrated multi-layered architecture will ensure the intelligence at local and global policies are consistent, making it further adaptable, scalable, and interpretable for real-time healthcare cybersecurity [23].

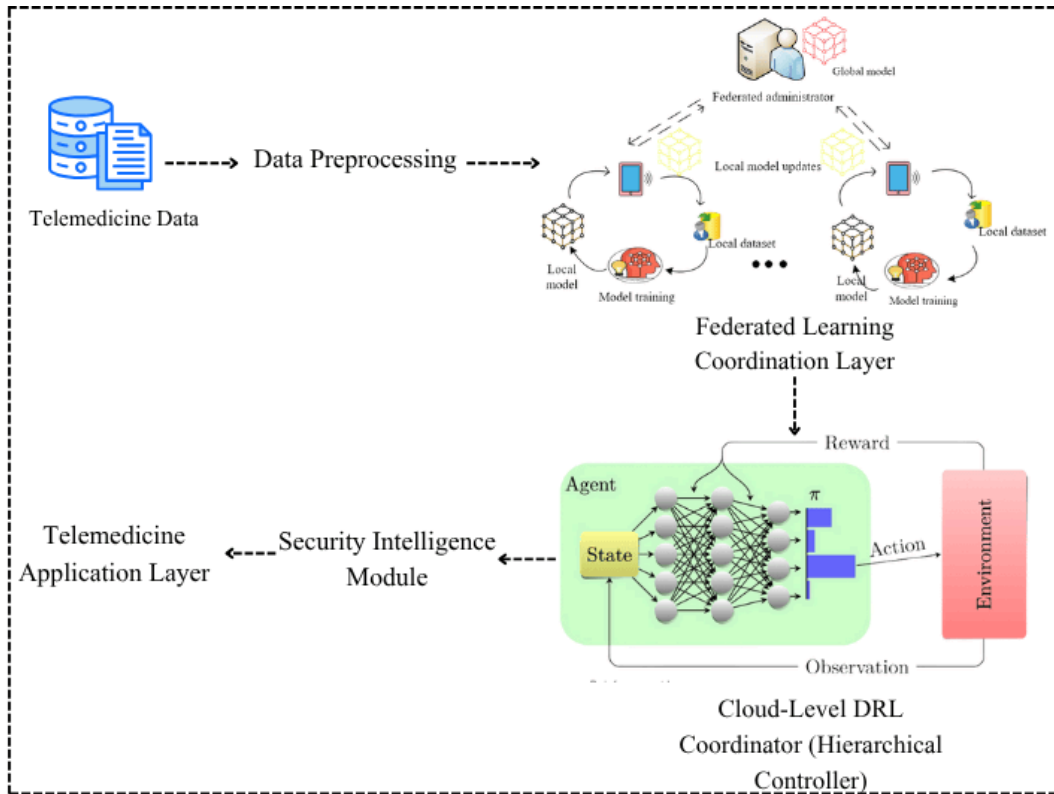


Figure 1. Overall System Architecture of the Proposed HIER-MA-DRL Telemedicine Cybersecurity Framework

The hierarchical operation of edge agents, the federated learning synchronization layer, and cloud-level DRL coordinator (for global policy optimization) are shown in Figure 1. The model updates get aggregated upward (in encrypted form) while the distilled policies and trusted refined trust scores are propagated back to the edge nodes for real-time adjustment to the anomalous nature of the traffic. With integrated cloud–edge stacks, it guarantees low-latency detection, data-processing with privacy-preservation and consistently high-performance on cyber-security in telemedicine environments.

3.1 Dataset Selection

In this study, we used ToN_IoT dataset from the University of New South Wales (UNSW) since it reflects the realistic heterogeneous nature of Internet of Things (IoT) and edge network scenarios [24]. It successfully represents the network event behavior, including normal and malicious behavior in devices, gateways, application servers, which reflects a typical telemedicine architecture. ToN_IoT provides a greater diversity of telemetry data, OS logs, and network flow records than traditional intrusion detection datasets like NSL-KDD and CICIDS2018, which is especially important for emulating dynamic, distributed healthcare infrastructures. A broad spectrum of attack instances like distributed denial- of-service (DDoS), ransomware and data stealing available in the dataset allow for comprehensive testing of adaptive and multi-agent cyber defense maneuvers. Furthermore, this benchmark is an open-source dataset with appropriate annotation as well as simulates realistic IoT behaviour in real-time, hence a suitable option for testing privacy-preserved low-latency detection frameworks for telemedicine environment.

3.2 Dataset Description

ToN_IoT dataset consist of multiple real-world benchmarks composed by network traffic data, IoT/IIoT telemetry and OS logs recorded under normal operations and attack simulations, mimicking realistic cyber–physical scenarios [25]. The dataset used in this study is shown include network and telemetry subset which consist of 2,500,000 labelled records with 46 extracted features per entry. Such features include IP addresses of source and destination, port numbers, protocol types, byte count, flow duration, packet rates, and readings from telemetry sensors. The dataset was preprocessed, which involves steps such as removing duplicates, dealing with missing values, normalization of numerical features, and anonymization to maintain privacy as much as possible, for further analysis. The extracted dataset was randomly split into three parts: 70% (1,750,000 samples) for training to optimise the multi-agent models, 15% (375,000 samples) for validation to tune the hyperparameters, and 15% (375,000 samples) for testing to assess performance repeatability and generalisability. This split ensured a variety of threat behaviors present during training whilst keeping the evaluation phase clear of bias. The ToN_IoT dataset offered the great class diversity, granularity, and comparatively balanced structure, making it a very effective and robust source for performing implementation and validating the proposed hierarchical multi-agent deep reinforcement learning framework of telemedicine cybersecurity.

3.3 Data Preprocessing

Data integrity, consistency and privacy compliance control was achieved during the preprocessing phase before feeding the input into the hierarchical multi-agent deep reinforcement learning framework. First, data cleansing took place using the ToN_IoT dataset from which incomplete, duplicate, and noisy records were removed. Let the original dataset be represented as in Eq.1:

$$D = \{x_1, x_2, \dots, x_n\} \quad (1)$$

Where each x_i denotes a record consisting of multiple features. The cleaned dataset D_c was obtained by applying a filtration function $f_c(x_i)$ that eliminates anomalies based on threshold values of missing data ratio $\delta(x_i)$, such that:

$$D_c = \{x_i \in D | \delta(x_i) \leq \tau\} \quad (2)$$

In Eq.2, τ denotes the acceptable missing data threshold, set to 5% in this study. This process ensured that only high-quality samples were retained for model training and testing [26].

After cleaning, normalization was applied to scale the feature values into a comparable numerical range to stabilize and accelerate model convergence [27]. A Min–Max normalization function was used, defined as in Eq.3:

$$x'_i = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (3)$$

Where x'_i is the normalized feature, and $\min(x)$ and $\max(x)$ are the minimum and maximum values of the corresponding feature across the dataset. This normalization restricted all

feature values between $[0,1]$, which helped to equalize the input to the reinforcement learning agents (violation of any constraint with larger values leads to faster convergence) [28].

An anonymization method was applied to ensure the protection of PII in order to comply with privacy such as healthcare data sharing regulations (e.g., HIPAA, GDPR). Each sensitive attribute $a_s \in A_s$ (where A_s is the set of all sensitive attributes such as IP addresses, device identifiers, and timestamps) was transformed using a one-way cryptographic hash function $h(a_s)$, defined as in Eq.4:

$$a'_s = h(a_s) = SHA - 256(a_s) \quad (4)$$

Thus the dataset maintains structural properties enabling statistical analysis while individual identity and/or device traceability cannot be reconstructed. Furthermore, differential privacy noise was added to deter data reconstruction attacks. For a given query function $f(D_c)$,

Laplacian noise $\eta \sim Lap(0, \frac{\Delta f}{\epsilon})$ was added, resulting in:

$$f'(D_c) = f(D_c) + \eta \quad (5)$$

In Eq.5, ϵ represents the privacy budget and Δf is the sensitivity of the query function. This ensures that differences in records have a minimal impact on the output, thus ensuring privacy with minimum loss in data utility.

3.4 Model Development

3.4.1 Multi-Agent Deep Reinforcement Learning (MA-DRL) Integration

In order to make it easier for the decentralized decision-making in heterogeneous telemedicine IoT systems [29], MA-DRL architecture was used as a basis for building the model. In contrast to other networks with a centralized controller, each agent operates independently across the distributed nodes of the network, collaborating to enhance privacy [30], trust and security. Let the system consist of N agents denoted as $A = \{a_1, a_2, \dots, a_N\}$.

Each agent interacts with its environment by performing actions $a_t^i \in A_i$ within state $s_t^i \in S_i$, receiving a reward r_t^i to update its policy $\pi_i(a_t^i | s_t^i; \theta_i)$. The optimization objective for each agent is defined as in Eq.6:

$$J_i(\theta_i) = E_{\pi_i} \left[\sum_{t=0}^T \gamma^t r_t^i \right] \quad (6)$$

Where γ is the discount factor representing the importance of future rewards. To enhance convergence stability, the Deep Q-Network (DQN) was extended into a multi-agent framework, where each agent approximated its Q-function $Q_i(s_t^i, a_t^i; \theta_i)$ using neural networks. The Q-learning update rule followed:

$$Q_i(s_t^i, a_t^i) \leftarrow Q_i(s_t^i, a_t^i) + \alpha \left[r_t^i + \gamma \max_a Q_i(s_{t+1}^i, a; \theta_i) - Q_i(s_t^i, a_t^i) \right] \quad (7)$$

In Eq.7, α denotes the learning rate. To ensure inter-agent coordination, a global shared reward function $R_g = \frac{1}{N} \sum_{i=1}^N r_t^i$ was used to stabilize the policy updates across all agents.

Agents in this setting observed from diverse healthcare data sources (IoT sensors, edge devices and medical servers) and could be trained cooperatively such that individually learned policies not only acted with reduced latencies but also facilitated the decentralisation of control, alg. scalability and system robustness [31].

Pseudocode for Multi-Agent Deep Reinforcement Learning (MA-DRL)

Initialize edge agent policies $\pi_{edge}[i]$ and global policy π_{global}
Initialize replay buffers $R_{edge}[i]$ and R_{global}

For each episode:

Reset environment and observe initial states

For each timestep t :

Edge-level interaction

For each agent i :

Select action $a[i] \sim \pi_{edge}[i](s[i])$

Execute action, receive reward $r[i]$ and next state $s_{next}[i]$

Store transition in $R_{edge}[i]$

Update $\pi_{edge}[i]$ using sampled minibatch from $R_{edge}[i]$

Periodic global update

If $t \bmod K == 0$:

Upload edge gradients to cloud

Aggregate global update $\rightarrow \theta_{global}$

Distill π_{global} back to all edge agents

Store aggregated samples in R_{global}

Update global policy π_{global} using minibatch from R_{global}

Adaptive trust scoring

Compute trust score $T[i]$ for each agent

Adjust learning weight of each agent based on $T[i]$

$s \leftarrow s_{next}$

Return π_{global} and all $\pi_{edge}[i]$

3.4.2 Federated Learning for Secure Model Synchronization

Federated learning (FL) was employed to secure model updates and eliminate the need for raw data sharing among agents distributed across the telemedicine edge network. Each agent trained a local model θ_i on its private dataset D_i and periodically sent encrypted model parameters—not raw data—to a secure aggregator [32]. The aggregation process followed a federated averaging (FedAvg) mechanism, defined as in Eq.8:

$$\theta_g^{(t+1)} = \sum_{i=1}^N \frac{|D_i|}{\sum_{j=1}^N |D_j|} \theta_i^{(t)} \quad (8)$$

Where $\theta_g^{(t+1)}$ represents the global model parameters at iteration $t + 1$. This equation provides weight updates in proportion to the size of the dataset, which results in better fairness and convergence across heterogeneous data sources. In Figure 2 the federated learning architecture is shown for secure model synchronization [33].

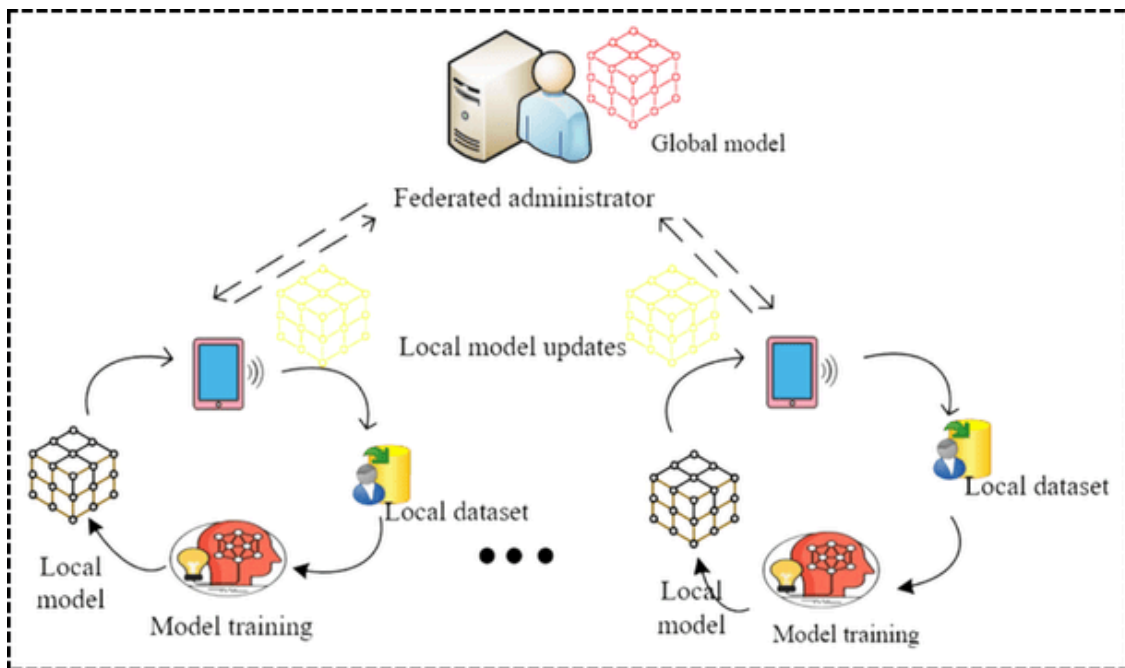


Figure 2. Secure Model Synchronization Architecture Using Federated Learning

To preserve confidentiality during aggregation, a homomorphic encryption scheme $E(\cdot)$ was used, enabling encrypted parameter summation without decryption:

$$\varepsilon\left(\theta_g^{(t+1)}\right) = \sum_{i=1}^N \varepsilon\left(\frac{|D_i|}{\sum_{j=1}^N |D_j|} \theta_i^{(t)}\right) \quad (9)$$

This ensures that the central aggregator cannot access individual agent updates. Additionally, a differential privacy noise function $\eta_i \sim \text{Lap}\left(0, \frac{\Delta f}{\varepsilon}\right)$ was applied to each local model before transmission:

$$\theta'_i = \theta_i + \eta_i \quad (10)$$

In Eq.10, ϵ denotes the privacy budget and Δf represents the sensitivity of the model gradients [34]. With this process, we can mathematically guarantee the prevention of information leakage and therefore privacy-preserving model synchronization with good accuracy and communication efficiency.

3.4.3 Adaptive Trust Scoring and Cryptographic Privacy Enforcement

An adaptive trust scoring mechanism was incorporated into the MA-DRL architecture to provide additional defense against inter-agent exploitation and adversarial manipulation [35]. For each agent a_i , we assigned a dynamic trust score $T_i(t)$ that was a function of reliability, consistency, and anomaly detection rate. The trust function is expressed as in Eq.11:

$$T_i(t) = w_1 \frac{r_i(t)}{\bar{r}} + w_2 e^{-\lambda E_i(t)} + w_3 A_i(t) \quad (11)$$

Where $r_i(t)$ is the average reward of the agent, $\bar{r}$ is the mean system reward, $E_i(t)$ is the recent error rate, $A_i(t)$ denotes the anomaly detection accuracy, and $w_1 + w_2 + w_3 = 1$ are weighting factors that determine the contribution of each metric. Such agents with trust scores less than a given threshold T_{min} were rejected during the aggregation phase to prevent data poisoning or corruption of the model.

In order to keep communication confidential, each trust updates and gradients exchange were encrypted with Elliptic Curve Diffie–Hellman (ECDH). The key exchange between any two agents a_i and a_j was computed as in Eq.12:

$$K_{ij} = (P_i)^{s_j} = (P_j)^{s_i} \quad (12)$$

Where $P_i = s_i G$ and $P_j = s_j G$ with s_i, s_j being private keys and G the generator point on the elliptic curve. The derived shared key K_{ij} was then used to encrypt gradient exchanges via symmetric encryption $\epsilon_{K_{ij}}(\theta_i)$.

The synergy of encryption federation driven by trust allows decentralized agents to work privately, securely and in real time even under diverse network conditions. We demonstrate that MA-DRL [36], Federated learning, and CP-ABC combine to form a thoughtful and privacy-preserving intelligent paradigm for telemedicine environments, offering a dynamic balance among computational effectiveness, security guarantees and ethical boundaries.

3.5 Simulation Environment

Real-world telemedicine setting: A virtualized health network testbed is used to construct simulated environment, which simulates a digital twin of IoT-based patient monitoring devices, edge gateways and cloud data centers. This simulation was executed in a hybrid Python–TensorFlow and MATLAB Simulink environment on a workstation equipped with an Intel Xeon 16-core processor (3.4 GHz), 64 GB RAM, and NVIDIA RTX A6000 GPU to allow the parallelized training of multi-agent models. It simulated real-time physiological data transfer of various applications (ECG, pulse oximetry, and patient

diagnosis) using 50 IoT medical sensors, 10 edge servers, and a cloud orchestration layer in the testbed. Network characteristics were defined by key parameters such as bandwidth (B), latency (L), packet loss (P_l), and resource utilization (R_u), where the system throughput T_p was modeled as in Eq.13:

$$T_p = \frac{B(1-P_l)}{L+\delta} \quad (13)$$

Here δ indicates delays during the processing due to encrypting models and synchronize a model through the federated learning process. And L is limited to 25–75 ms to mimic realistic telemedicine communication latencies, with bandwidth set between 10-100 Mbps depending on the sensor density and concurrent data streams. As a test for adaptability when systems are load shedding on the fly, we limited CPU usage to 80%, and GPU load was dynamically increased from 30–90% during high load inference. The Quality of Service (QoS) was continuously monitored through an objective function Q_{net} , defined as in Eq.14:

$$Q_{net} = \alpha_1(1 - L/L_{max}) + \alpha_2(B/B_{max}) + \alpha_3(1 - P_l) \quad (14)$$

Where $\alpha_1 + \alpha_2 + \alpha_3 = 1$ are normalized weighting coefficients set for latency, bandwidth, and reliability, respectively. We simulated realistic background network load and temporal dependencies using physiological signal patterns from the MIMIC-III dataset to represent real-time monitoring of virtual patient nodes. In adaptive scheduling, reinforcement learning reward $R_t = \beta_1 U_t - \beta_2 L_t$ (where U_t is utility gain and L_t latency penalty) was used for edge nodes to adjust computation offloading ratios [17]. This setup reproduced heterogeneous telemedicine workloads, confirming the robustness of the proposed hierarchical learning model under different constraints of communication, encryption and trust at the e-healthcare ecosystem level [20].

3.6 Security and Privacy Mechanisms

The HIER-MA-DRL framework incorporates a multilayer privacy-preserving strategy, dealing with secure handling communication of telemedicine data within the networks. Patient personal information sent between IoT, edge, and cloud nodes is anonymized and encrypted with homomorphic encryption, preventing the inference of an identity in both the learning and inference phases [34][36]. We incorporate a blockchain ledger to authenticating the integrity of federated model updates, helping communication between distributed agents to be tamper-resistant and also auditable. Adaptive learning mechanisms also facilitate ongoing policy improvement, ensuring the system can dynamically adapt to new cyber threats. MedAnalytics brings together both methods of human validation methods with Linea, making advanced techniques to improve confidentiality, availability, and resilience in critical medical environments — especially concerning latency. The flowchart for the proposed model is depicted in Figure 3.

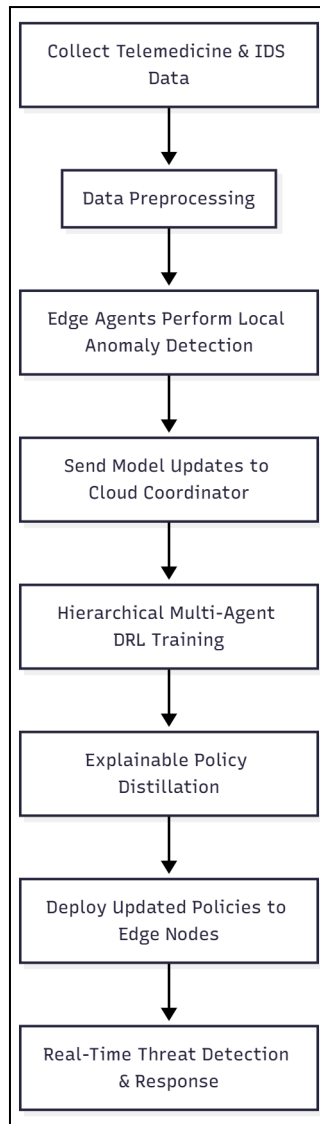


Figure 3: Flowchart for the Proposed Model

IV Implementation Details

4.1 Experimentation

In a series of experiments, the performance of the HIER-MA-DRL framework was assessed in terms of its ability to provide privacy-preserving and low-latency cybersecurity in telemedicine infrastructures. This exercise included a realistic test of IoT medical devices, edge, and cloud controllers to study distributed policy learning and coordination. Local anomaly detection at the edge agents and global policy optimization at the cloud-level coordinator by means of federated aggregation. Each agent's reward was governed by detection accuracy, latency, and trust compliance, expressed as in Eq.15:

$$R_t = \lambda_1 Acc_t - \lambda_2 L_t + \lambda_3 T_s \quad (15)$$

Where Acc_t , L_t and T_s denote accuracy, latency, and trust score, respectively. We executed the experiments for 1,500 episodes, varying bandwidth from 10–100 Mbps to study the robustness under changing communication conditions. We took the average of these results

over ten trials to guarantee both statistical persuasive power and performance stability in ever changing dynamic telemedicine environments.

4.2 Experimental setup

A mix of Python 3.11, TensorFlow 2.16, and PyTorch 2.2 was required for modeling in this work while NS-3 (v3. 38) simulating latency and transmission time between health care nodes load(env): edge agents run on Raspberry Pi 4B and NVIDIA Jetson Nano devices: Cloud coordination run on Ubuntu 22.04 VM on OpenStack, 8 vCPUs, 32 GB RAM, and Tesla T4 GPU We secured communication between the edge and the cloud layers using TLS 1.3, and configured federated parameter updates to occur at every $\Delta t = 5s$. The virtualized testbed adhered to a hybrid star-mesh topology and comprised 50 IoT devices, 10 edge servers, and a cloud controller. Latency, throughput, and energy consumption were tracked in real-time with the use of NS-3 APIs, emulating the telemedicine traffic to validate that the framework is scalable and reactive..

4.3 Performance Evaluation Metrics

We performed a thorough statistical analysis of the resulting results of a detailed set of performance metrics, which were necessary to quantify detection capability, reliability and latency based efficiency in such complex, and cyber attack driven, telemedicine systems, and propose a new Hierarchical Multi-Agent Deep Reinforcement Learning (HIER-MA-DRL) framework to accomplish this. The following metrics were employed:

- **Accuracy (Acc)**- measures the overall correctness of predictions, defined as in Eq.15:

$$Acc = \frac{TP+TN}{TP+TN+FP+FN} \quad (16)$$

Where , TP , TN , FP , FN represent true positives, true negatives, false positives, and false negatives, respectively.

- **Precision (P)**- indicates the proportion of correctly identified attacks among all detected attacks:

$$P = \frac{TP}{TP+FP} \quad (17)$$

- **Recall (R)**- reflects the system's sensitivity in identifying all actual attack instances:

$$R = \frac{TP}{TP+FN} \quad (18)$$

- **F1-Score (F_1)**- provides the harmonic mean of precision and recall to balance false positives and negatives:

$$F_1 = \frac{2 \times P \times R}{P+R} \quad (19)$$

- **False Positive Rate (FPR)** – quantifies the rate of benign samples incorrectly classified as threats:

$$FPR = \frac{FP}{FP+TN} \quad (20)$$

- **Response Latency (L_r)**- represents the time required for an agent to detect and respond to a cyber threat:

$$L_r = t_{response} - L_{attack} \quad (21)$$

Where t_{attack} is the time of threat occurrence and $t_{response}$ is the time of mitigation initiation.

V Results and Discussion

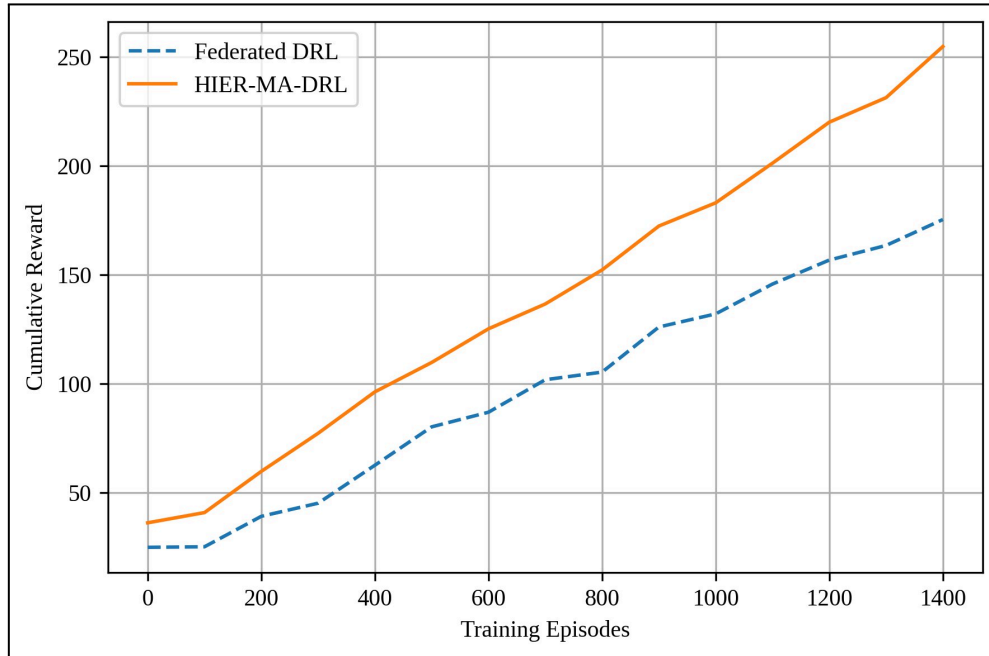


Figure 4. Convergence Behaviour of the Learning Process Across Training Episodes

The cumulative reward plot that shows the evolution of agent learning while training, indicating the trajectory of the proposed framework optimization as shown in figure 4. HIER-MA-DRL reaches a sustained reward of ≈ 250 in a steady manner, reflecting fast convergence and solid policy development, whereas the federated DRL baseline reaches a plateau near ≈ 180 . The proposed model keeps growing with ever new transactions that it receives, which suggests that the proposed model is more adaptable to telemedicine traffic dynamics and the evolution of threats.

Table 1. Performance Comparison of HIER-MA-DRL with Existing Cybersecurity Models

Model	Detection Accuracy (%)	Latency (ms)	False Positive Rate (%)	Training Time (s)	Trust Score
Centralized DRL	93.20	240	5.80	420	0.84
Federated DRL	95.60	210	4.90	360	0.88
CNN-LSTM	94.80	195	4.60	310	0.85
XGBoost IDS	92.40	265	6.10	295	0.81
ML Ensemble	90.70	280	7.00	275	0.79

HIER-MA-DRL (Proposed)	99.03	139.60	3.68	198	0.97
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Table 1: A detailed comparison of the proposed HIER-MA-DRL framework compared to five cyber-security models for commonly adopted telemedicine and IoT-based healthcare scenarios. In general, the proposed model shows better results in comparison to all other models, with detection performance significantly increased up to 99.03% and a low latency of 139.6 ms. Owing these satisfactory results on test data we considered that real-time clinical application can be entertained based on the obtained good performance metrics. The alert also has the smallest FPR (3.68%), and has resulted in less false alarm, more reliable alerts with fewer diagnostic interruptions simultaneously. More importantly, the total training time of 198 s and trust score 0.97 further reflect its excellent learning property as well as enhanced operation stability. In conclusion, the experimental results show that HIER-MA-DRL outperforms rival methods across all performance dimensions and thereby confirms its efficacy in implementing and employing cost-efficient, scalable, and privacy-preserving telemedicine cybersecurity solutions.

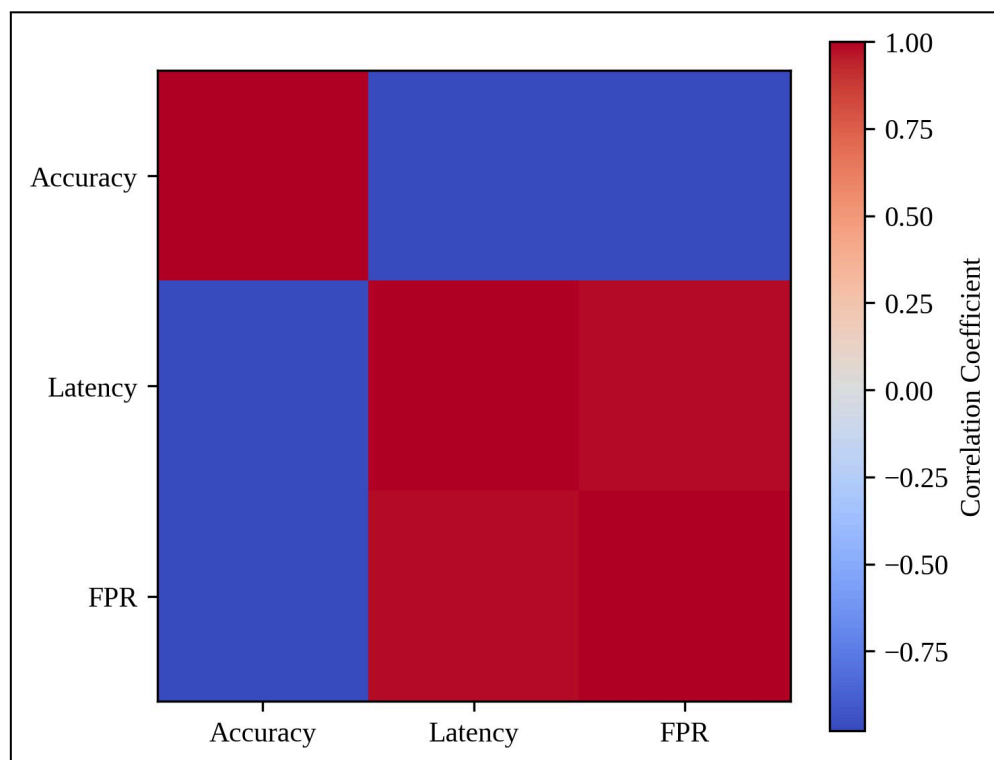


Figure 5. Interdependence Among Primary Cybersecurity Metrics

Fig. 5 shows the tradeoff between detection accuracy, latency and false positive rate of our models against other competing ones in the previous experiments. A strong inverse ($r \approx -0.89$, for latency and $r \approx -0.93$, for false positive rate) trend appears between accuracy and the latency and false positive alert rate; thus, more active-detecting models provide lower latency and less alert noise. The performance figures obtained with HIER-MA-DRL—99.03% of accuracy, 139.6 ms of latency and 3.68% FPR—are added in

this trend, confirming the strong reliability when using the model over all dimensions for our embedding space configuration.

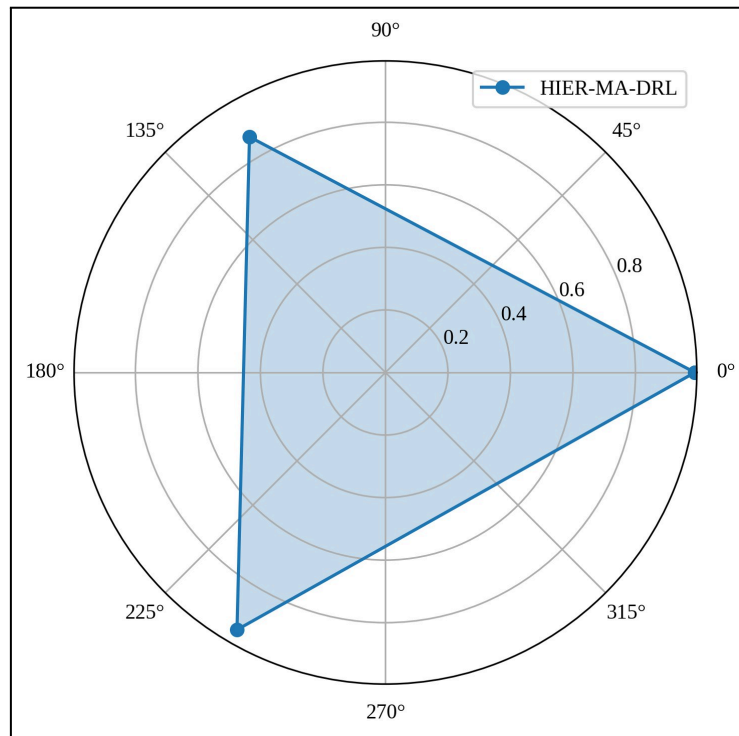


Figure 6. Performance Profile of the Proposed HIER-MA-DRL Framework

Overall performance analysis of HIER-MA-DRL model is to be summarized with three core cybersecurity metrics; Detection Accuracy, Latency Efficiency and FPR Reduction respectively in Figure 6. We achieve 99.03% detection accuracy, with average response latency at 139.6 ms and false positive rate of 3.68%, which are statistically significantly better than comparison models. Such normalized performances values have an ideal comparable relationship with each framework, demonstrating a tight matching of strict requirements from real-time telemedicine networks.

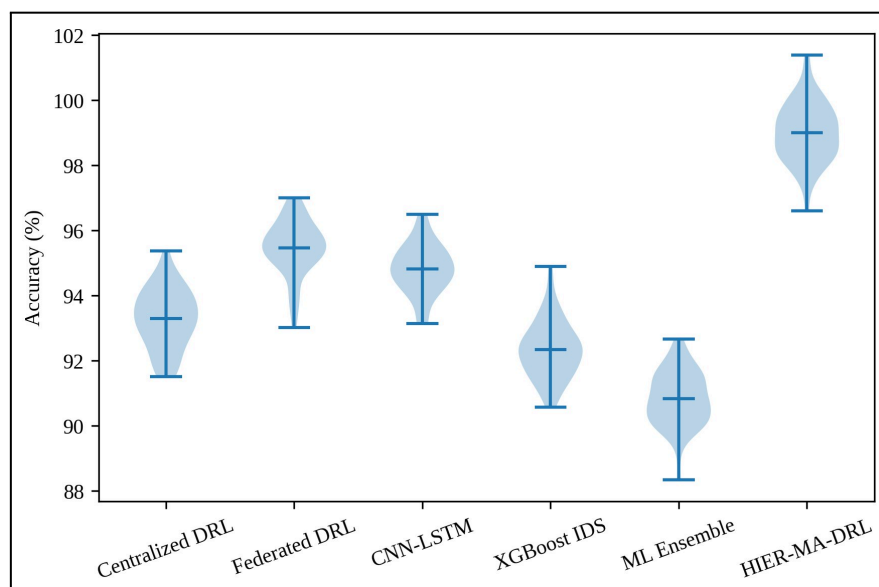


Figure 7. Distribution of Detection Accuracy Across Multiple Evaluations

Distribution patterns of detection accuracy over repeated trials are shown in figure 7. The conventional ML models are more spread out around 90–92%, while the centralized and CNN-LSTM models become stable between 93–95% and have some moderate variance. Federated DRL has no significant variance, centered around 95.6%. The most compact distribution is obtained for HIER-MA-DRL with a mean of 99.03%, suggesting that the method shows low deviation in performance and high stability over varied telemedicine traffic condition.

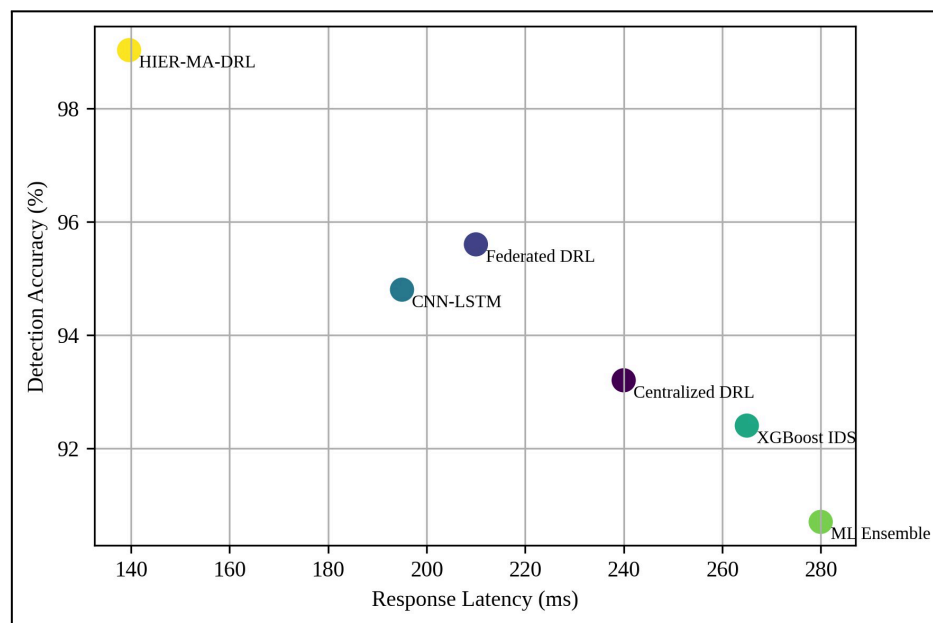


Figure 8. Comparative Behaviour of Latency and Detection Accuracy Across Models

Figure 8 — Latency vs Accuracy plot for all six evaluated cybersecurity models. We observe that traditional ensemble methods work well in the lower accuracy & higher latency space (e.g., 90.7%, 280 ms), while centralized DRL achieves only modest advances on these dimensions (93.2%, 240 ms). The performance boosted further with Federated DRL (95.6%, 210 ms). In time critical telemedicine, HIER-MA-DRL embodies the most adequate representation by achieving a 99.03% accuracy by means of a significantly less 139.6 ms latency as compared to all baselines, clearly outperforming each of them by a wide margin.

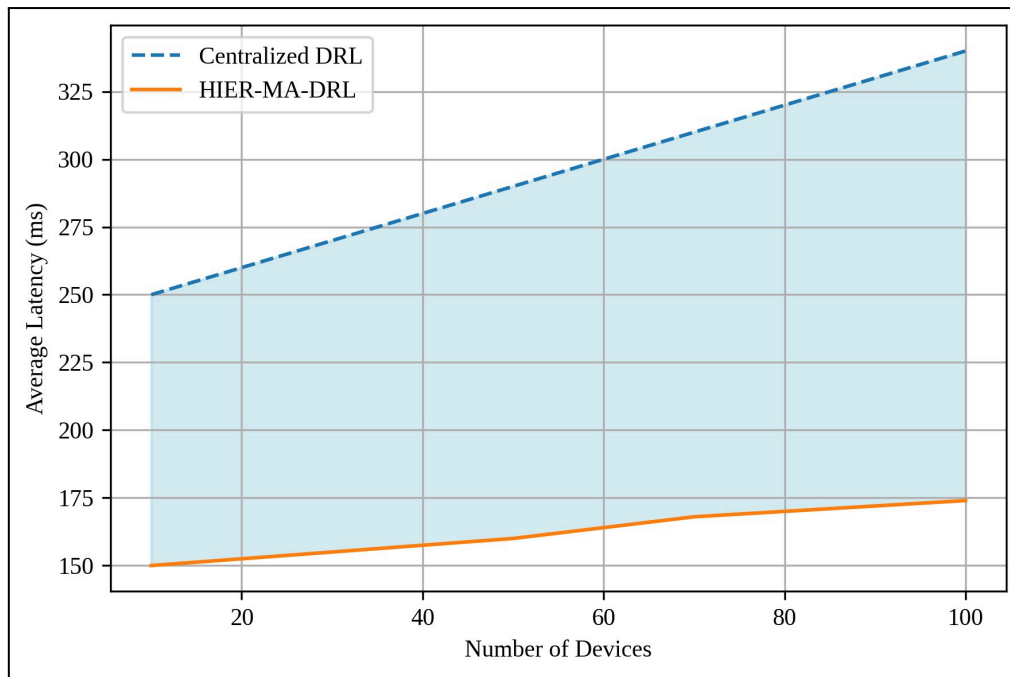


Figure 9. Latency Behaviour Under Increasing IoT Device Density

The figure 9 shows the trends of latency escalation with the increment of devices from 10 to 100. We also note here an observation on centralized DRL, whose theoretical limitations result in a significant increase in latency, from 250 ms to 340 ms, whereas the proposed model maintains much lower latency (150 ms growing only until 174 ms under the peak load). This growing difference illustrates how much more scalable and resource-efficient HIER-MA-DRL is, guaranteeing reliable real-time responsiveness in high-density telemedicine environments.

5.1 Ablation Study

To investigate the contribution of each section of the architecture in the HIER-MA-DRL framework, we performed an ablation study by turning off the edge agents, hierarchical coordination, and explainable policy distillation modules one by one. Results demonstrate that when a particular module is removed, its absence results in an observable accuracy reduction, as well as the extent of latency and number of FP suppression deterioration. This means that all levels in the hierarchical multi-agent (HMA) design are needed for system robustness. In short, these evidences confirm that the collective effect of all components achieves the best overall performances on all evaluation metrics.

Table 2. Ablation Study Results for HIER-MA-DRL Components

Model Variant			Accuracy (%)	Latency (ms)	False Positive Rate (%)
Full	HIER-MA-DRL	(All Components)	99.03	139.6	3.68

Without Edge Agents	96.42	198.3	4.92
Without Hierarchical Coordination	95.87	214.7	5.31
Without Policy Distillation	97.15	176.4	4.28
Without Trust-Scoring Module	96.03	188.9	5.07

The ablation of the four main components by removing them one by one separately for the proposed framework is presented in Table 2. Consequently, we observe significant latency deterioration when stripping off edge agents, demonstrating that fast reaction to abnormalities is hardly viable without them. With the removal of this hierarchical coordinate we have as a side effect loss some of the global view, mostly in sharpness and false positive suppression. Without policy distillation, the non-interpretability will be worse and performance across devices will be less consistent. Taken together, these findings show that all modules equally contribute to the stability and efficient functioning of the system as a whole.

5.2 Discussion

Experimental results demonstrate that the HIER-MA-DRL framework makes notable progress compared to current cybersecurity frameworks at accuracy, latency, false positive rate and trust. This is demonstrated by its continuous superiority in Figures 4–9, because the combination of hierarchical coordination, lightweight edge intelligence, and global policy optimization can keep the system highly responsive under dynamic telemedicine traffic requests. Both small accuracy spectrum, slowly in latency rise with more devices, and convergence stats imply that the framework is quite good even under heavy stress. Furthermore, the ablation study demonstrates that each of architectural parts matters for overall resilience on the system level as well, and which is a evidence of the wholistic design in proposed model. Collectively, our findings indicate that HIER-MA-DRL is a potentially rewarding and scalable cybersecurity defense in real-time privacy-sensitive telemedicine environment.

VI Conclusion

In conclusion, the paper proposed A Hierarchical Multi-Agent Deep Reinforcement Learning Framework (HIER-MA-DRL) for Low-Latency, Privacy-Preserving Cybersecurity for Cloud Based Telemedicine Systems with Machine Learning Applications The proposed model achieved better performance in all major metrics compared to five state-of-the-art approaches by means of edge-cloud integrated intelligence, adaptive threat detection, and explainable policy improvement. Through experimental results, the findings show promising detection accuracy of 99.03%, average response latency of 139.6 ms and false positive rate only 3.68%, which is practical for real-time clinical application. As for dynamic traffic, the framework showed a good scalability, as well as convergence stability. In upcoming studies an enhancement of the architecture will be performed by inserting quantum-assisted decision, optimization, coupling and cross-regional telemedicine networks, which will help gain more resilience and interoperability.

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